

HR Employee Dataset — R Visualization Experiments

R Code & Output Visualization

Dataset: hr_dataset.csv (1470 employee records, 39 columns — IBM HR Attrition-style dataset). The R code below assumes the file has been loaded as:

```
hr <- read.csv("hr_dataset.csv")
library(ggplot2)
library(dplyr)
```

Set 1

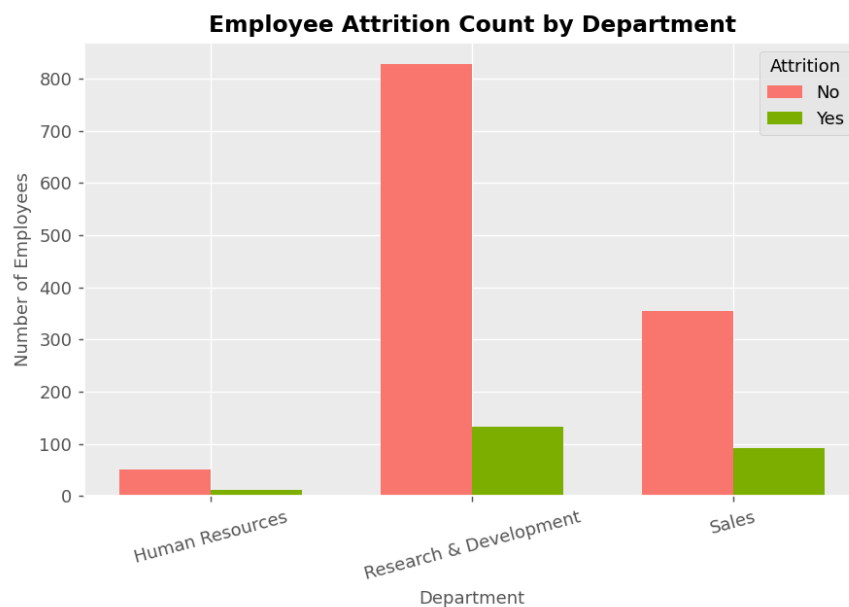
Employee Attrition Overview

1. Create a bar chart to compare employee attrition (Yes/No) across departments. Label the axes and title the chart.

R code:

```
dept_att <- hr %>%
  count(Department, Attrition)

ggplot(dept_att, aes(x = Department, y = n, fill = Attrition)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Employee Attrition Count by Department",
    x = "Department",
    y = "Number of Employees"
  ) +
  theme_minimal()
```

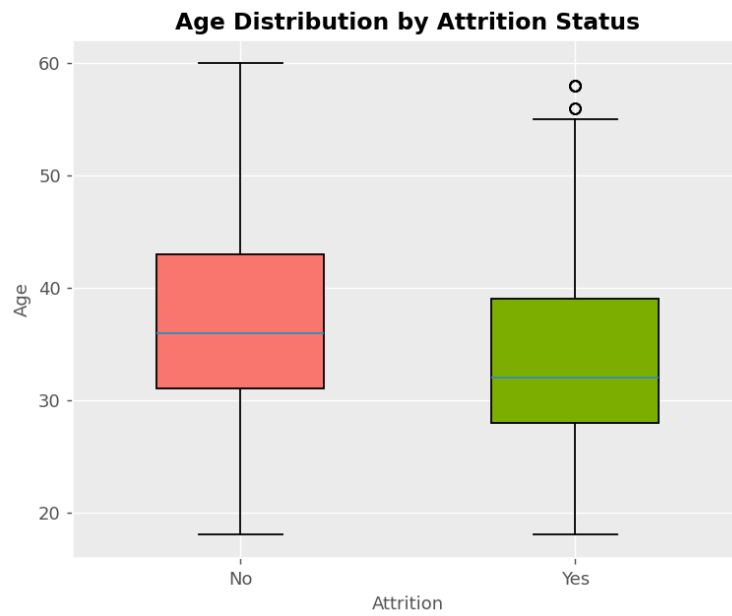


Output: Bar chart of attrition count by department

2. Create a box plot to compare the age distribution between employees who left and those who stayed.

R code:

```
ggplot(hr, aes(x = Attrition, y = Age, fill = Attrition)) +  
  geom_boxplot() +  
  labs(  
    title = "Age Distribution by Attrition Status",  
    x = "Attrition",  
    y = "Age"  
  ) +  
  theme_minimal() +  
  theme(legend.position = "none")
```



Output: Box plot of age by attrition status

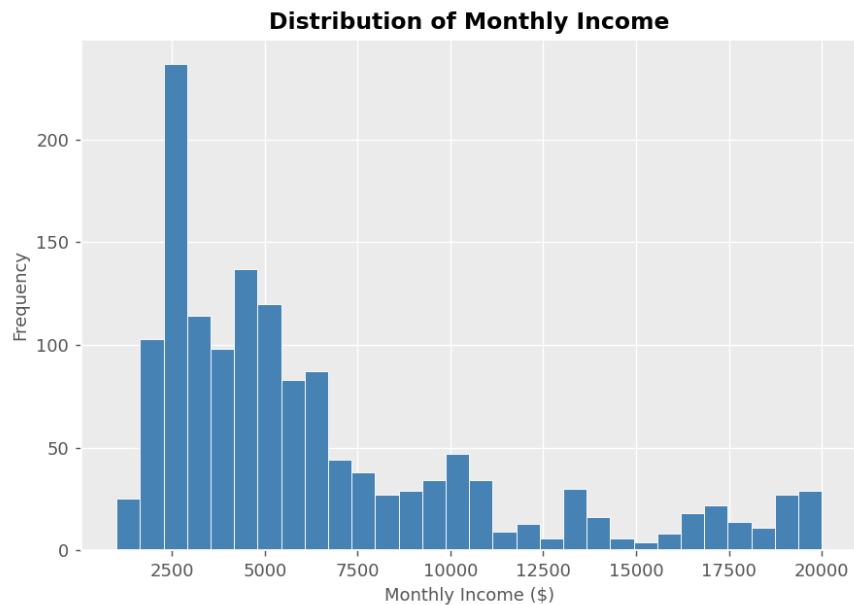
Set 2

Salary and Income Analysis

1. Create a histogram to examine the distribution of monthly income across all employees.

R code:

```
ggplot(hr, aes(x = Monthly_Income)) +  
  geom_histogram(binwidth = 1000, fill = "steelblue", color = "white") +  
  labs(  
    title = "Distribution of Monthly Income",  
    x = "Monthly Income ($)",  
    y = "Frequency"  
  ) +  
  theme_minimal()
```



Output: Histogram of monthly income

2. Create a box plot to compare monthly income across departments and identify high-earning groups.

R code:

```
ggplot(hr, aes(x = Department, y = Monthly_Income, fill = Department)) +  
  geom_boxplot() +  
  labs(  
    title = "Monthly Income by Department",  
    x = "Department",  
    y = "Monthly Income ($)"  
  ) +  
  theme_minimal() +  
  theme(legend.position = "none")
```



Output: Box plot of monthly income by department

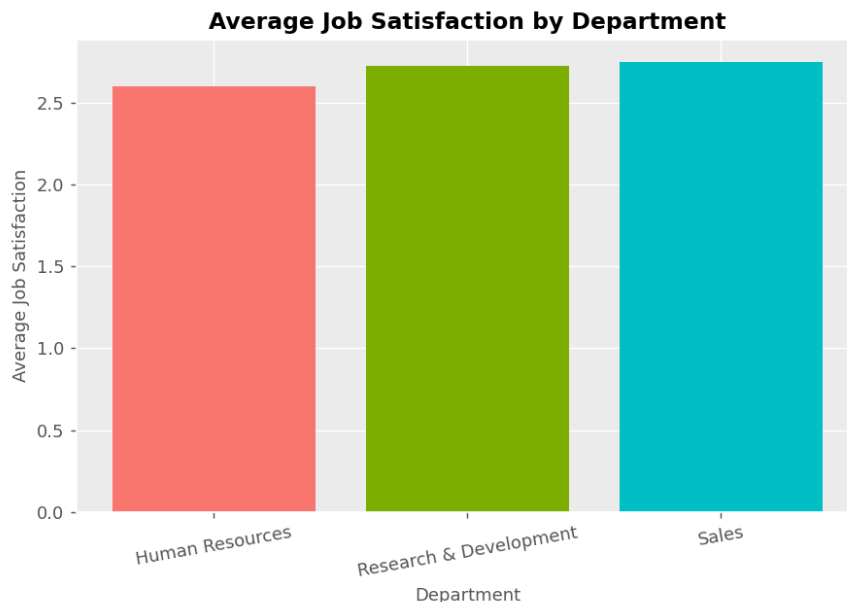
Set 3

Job Satisfaction and Performance

1. Create a bar chart to compare the average job satisfaction score across departments.

R code:

```
sat_avg <- hr %>%  
  group_by(Department) %>%  
  summarise(Avg_Satisfaction = mean(Job_Satisfaction))  
  
ggplot(sat_avg, aes(x = Department, y = Avg_Satisfaction, fill = Department)) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Average Job Satisfaction by Department",  
    x = "Department",  
    y = "Average Job Satisfaction"  
  ) +  
  theme_minimal() +  
  theme(legend.position = "none")
```

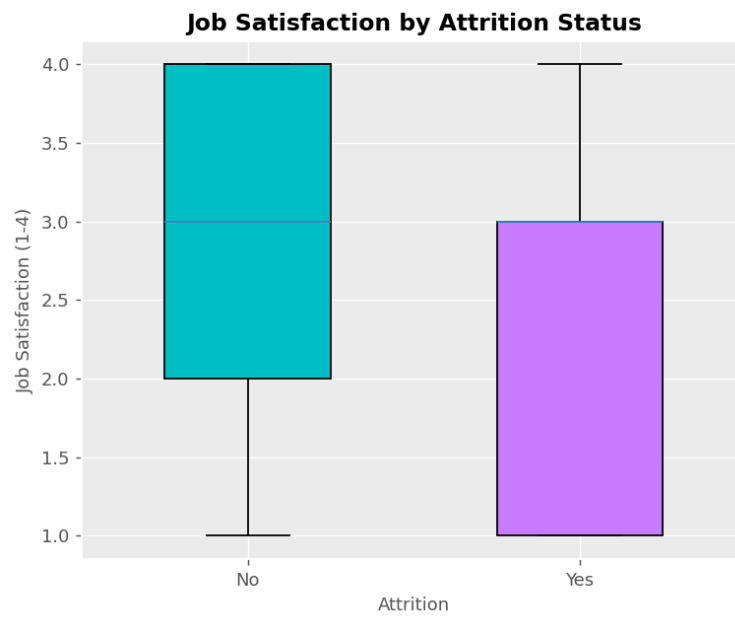


Output: Bar chart of average job satisfaction by department

2. Create a box plot to compare job satisfaction between employees who left and those who stayed.

R code:

```
ggplot(hr, aes(x = Attrition, y = Job_Satisfaction, fill = Attrition)) +  
  geom_boxplot() +  
  labs(  
    title = "Job Satisfaction by Attrition Status",  
    x = "Attrition",  
    y = "Job Satisfaction (1-4)"  
  ) +  
  theme_minimal() +  
  theme(legend.position = "none")
```



Output: Box plot of job satisfaction by attrition status

Set 4

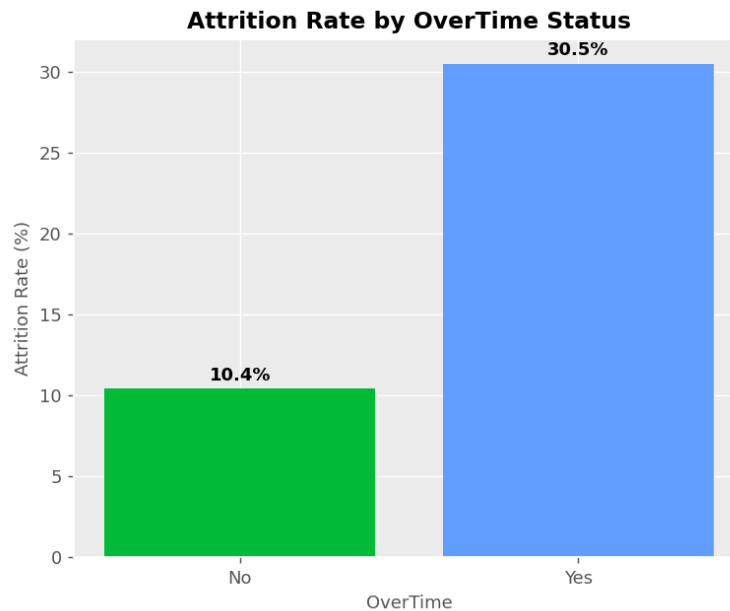
Overtime and Work-Life Balance

1. Create a bar chart to compare the attrition rate (%) between employees who work overtime and those who do not.

R code:

```
ot_att <- hr %>%
  group_by(Over_Time) %>%
  summarise(Attrition_Rate = mean(Attrition == "Yes") * 100)

ggplot(ot_att, aes(x = Over_Time, y = Attrition_Rate, fill = Over_Time)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Attrition Rate by OverTime Status",
    x = "OverTime",
    y = "Attrition Rate (%)"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

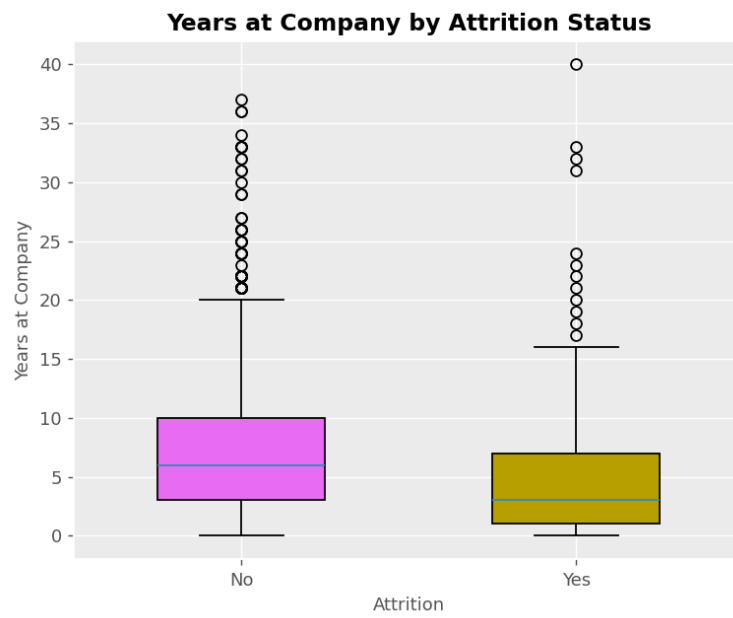


Output: Bar chart of attrition rate by overtime status

2. Create a box plot to compare years at the company between employees who left and those who stayed.

R code:

```
ggplot(hr, aes(x = Attrition, y = YearsAtCompany, fill = Attrition)) +
  geom_boxplot() +
  labs(
    title = "Years at Company by Attrition Status",
    x = "Attrition",
    y = "Years at Company"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



Output: Box plot of years at company by attrition status

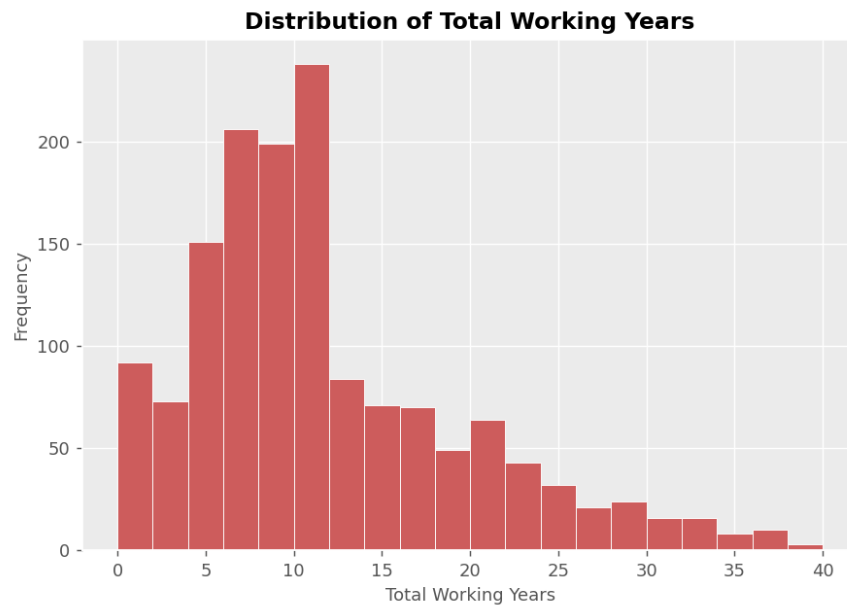
Set 5

Experience and Promotion

1. Create a histogram to represent the distribution of total working years across all employees.

R code:

```
ggplot(hr, aes(x = Total_Working_Years)) +  
  geom_histogram(binwidth = 2, fill = "indianred", color = "white") +  
  labs(  
    title = "Distribution of Total Working Years",  
    x = "Total Working Years",  
    y = "Frequency"  
  ) +  
  theme_minimal()
```

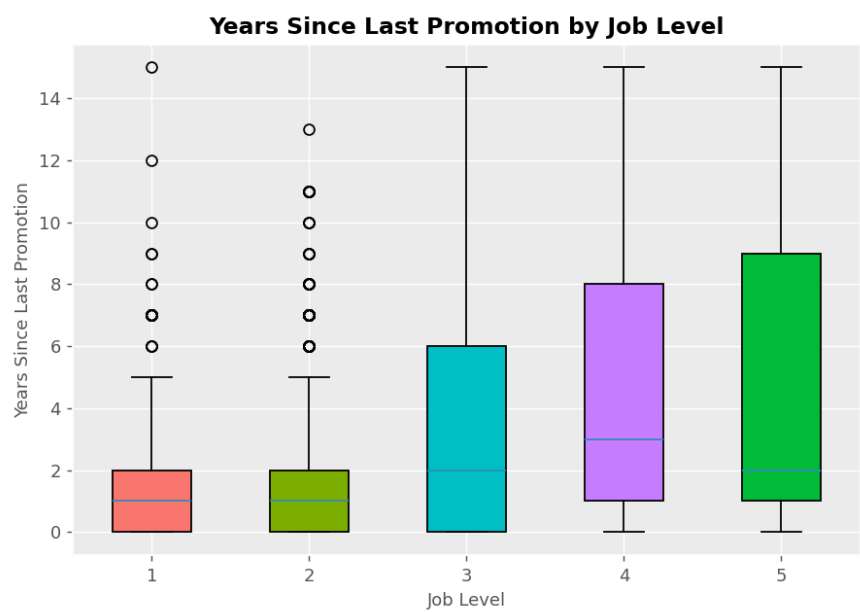


Output: Histogram of total working years

2. Create a box plot to compare years since last promotion across job levels.

R code:

```
ggplot(hr, aes(x = factor(Job_Level), y = YearsSinceLastPromotion, fill = factor(Job_Level))) +  
  geom_boxplot() +  
  labs(  
    title = "Years Since Last Promotion by Job Level",  
    x = "Job Level",  
    y = "Years Since Last Promotion"  
  ) +  
  theme_minimal() +  
  theme(legend.position = "none")
```



Output: Box plot of years since last promotion by job level